fn approximate_to_tradeoff

Yu-Ju Ku, Jordan Awan, Aishwarya Ramasethu, Michael Shoemate

December 2, 2024

1 Hoare Triple

Preconditions

Compiler-verified

- Argument `param` of type `(f64, f64)`

User-verified

None

Pseudocode

```
def appproximate_to_tradeoff(
    params: tuple[f64, f64]
) -> tuple[Callable[[RBig], RBig], RBig]:
    epsilon, delta = params

    exp_eps = epsilon.with_rounding(Down).exp()  #
    exp_eps = RBig.try_from(exp_eps)

    exp_neg_eps = (-epsilon).with_rounding(Up).exp()  #
    exp_neg_eps = RBig.try_from(exp_neg_eps)

    fixed_point = (RBig(1) - delta) / (RBig(1) + exp_eps)

    if fixed_point >= RBig(1, 2):
        raise ValueError("fixed point of tradeoff curve must be less than 1/2")

    def tradeoff(alpha: RBig) -> RBig:  #
        t1 = RBig(1) - delta - exp_eps * alpha
        t2 = exp_neg_eps * (RBig(1) - delta - alpha)
        return max(max(t1, t2), RBig(0))

    return tradeoff, fixed_point
```

Postcondition

Theorem 1.1. Given a pair of epsilon and delta, return the corresponding symmetric nontrivial f-DP tradeoff curve with conservative arithmetic, as well as the fixed point c where $c = f(c)$. Returns an error if epsilon or delta are invalid.

Proof.

Definition 1.2 (Awan and Vadhan 2023, Definition 2.2)**.** Let $\epsilon > 0$ and $\delta \geq 0$, and define

$$f_{\epsilon,\delta}(\alpha) = \max(0, 1 - \delta - \exp(\epsilon)\alpha, \exp(-\epsilon)(1 - \delta - \alpha)). \tag{1}$$

Then we say that a mechanism M satisfies (ϵ, δ)-DP if it satisfies $f_{\epsilon,\delta}$-DP.

On lines 6 and 9, the transcendental functions are computed in a manner which results in over-estimates of the privacy loss, and therefore a tradeoff curve that bows further away from $1 - x$. The results of these constants are converted exactly into rationals to be used in the tradeoff function. The function defined on line 17 implements the formula in 1.2 with exact fractional arithmetic.

The fixed-point c of the tradeoff function is $(1 - \delta)/(1 + e^{\epsilon})$, because $f_{\epsilon,\delta}(c) = c$.

$$\begin{aligned}
&f_{\epsilon,\delta}((1 - \delta)/(1 + e^{\epsilon})) && (2)\\
&= \max(0, 1 - \delta - \exp(\epsilon)((1 - \delta)/(1 + e^{\epsilon})), \exp(-\epsilon)(1 - \delta - ((1 - \delta)/(1 + e^{\epsilon})))) && (3)\\
&= \max(0, (1 - \delta)/(1 + e^{\epsilon}), (1 - \delta)/(1 + e^{\epsilon})) && (4)\\
&= (1 - \delta)/(1 + e^{\epsilon}) && (5)
\end{aligned}$$

□

References

Awan, Jordan and Salil Vadhan (2023). "Canonical Noise Distributions and Private Hypothesis Tests". In: *The Annals of Statistics* 51.2, pp. 547–572.